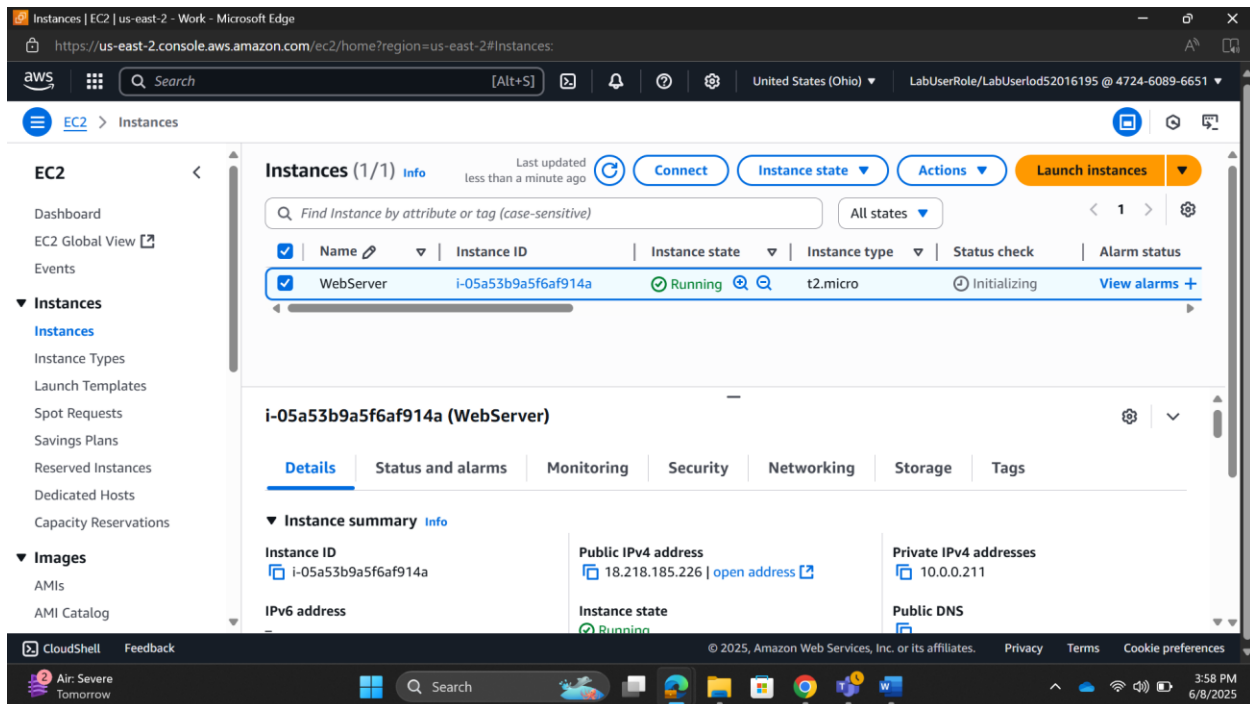


Lab 5 Create a CloudWatch Alarm

In this challenge, you will configure Amazon CloudWatch to generate an alarm when CPU utilization reaches a specific level. First, you will create a CloudWatch alarm to monitor CPU utilization on an Amazon Elastic Cloud Compute (Amazon EC2) instance, and then you will connect to the console of the EC2 instance by using EC2 Instance Connect. Next, you will install a stress testing tool, and then you will run a stress test to simulate load. Finally, you will monitor the CloudWatch alarm until it is triggered. Note: Once you begin the challenge lab, you will not be able to pause, save, or exit and then return to your challenge lab. Please ensure that you have set aside enough time to complete the challenge lab before you start.

Task 1: Connect to an EC2 instance



Check server running or not:

<http://ec2-18-218-185-226.us-east-2.compute.amazonaws.com>

10.0.0.211

PHP Version 5.4.16

System	Linux ip-10-0-0-211.us-east-2.compute.internal 4.14.355-277.647.amzn2.x86_64 #1 SMP Mon May 19 09:44:01 UTC 2025 x86_64
Build Date	Apr 7 2025 16:22:56
Server API	Apache 2.0 Handler
Virtual Directory Support	disabled
Configuration File (php.ini) Path	/etc
Loaded Configuration File	/etc/php.ini
Scan this dir for additional .ini files	/etc/php.d
Additional .ini files parsed	/etc/php.d/curl.ini, /etc/php.d/fileinfo.ini, /etc/php.d/json.ini, /etc/php.d/phar.ini, /etc/php.d/zip.ini
PHP API	20100412
PHP Extension	20100525
Zend Extension	220100525
Zend Extension Build	API220100525.NTS
PHP Extension Build	API20100525.NTS
Debug Build	no
Thread Safety	disabled
Zend Signal Handling	disabled
Zend Memory Manager	enabled

Task 2: Create a CloudWatch alarm to monitor CPU utilization

CloudWatch -> Alarms -> all alarms -> create alarm

Alarms | CloudWatch | us-east-2 - Work - Microsoft Edge

https://us-east-2.console.aws.amazon.com/cloudwatch/home?region=us-east-2#alarmsV2:

CloudWatch > Alarms

Alarms (0)

☐ Hide Auto Scaling alarms

Clear selection

Create composite alarm

Actions

Create alarm

Search

Alarm state: Any

Alarm type: Any

Actions status: Any

< 1 >

	Name	State	Last state update (UTC)	Conditions
	No alarms			
	No alarms to display			
	Read more about Alarms			
	Create alarm			

CloudShell Feedback

© 2025, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

EC2 -> select Per-Instance Metrics

Select metric | Alarms | CloudWatch | us-east-2 - Work - Microsoft Edge

https://us-east-2.console.aws.amazon.com/cloudwatch/home?region=us-east-2#alarmsV2:create?~(Page~'MetricSelection~AlarmType~'MetricAlarm~AlarmData~(Metrics~(~)~AlarmName...

Select metric

Untitled graph 

1h 3h 12h 1d 3d 1w Custom  UTC timezone  Line  

 Add math  Add query

Browse Multi source query Graphed metrics (1) Options Source

☐	WebServer	i-05a53b9a5f6af9...	NetworkPacketsOut 	No alarms
☐	WebServer	i-05a53b9a5f6af9...	MetadataNoToken 	No alarms
☒	WebServer	i-05a53b9a5f6af9...	CPUUtilization 	No alarms
☐	WebServer	i-05a53b9a5f6af9...	NetworkPacketsIn 	No alarms
☐	WebServer	i-05a53b9a5f6af9...	NetworkIn 	No alarms
☐	WebServer	i-05a53b9a5f6af9...	StatusCheckFailed_System 	No alarms

Cancel **Select metric**

Create alarm | Alarms | CloudWatch | us-east-2 - Work - Microsoft Edge

https://us-east-2.console.aws.amazon.com/cloudwatch/home?region=us-east-2#alarmsV2:create?~(Page~'MetricSelection~AlarmType~'MetricAlarm~AlarmData~(Namespace~'AWS'2fEC2...

CloudWatch > Alarms > Create alarm

Period
1 minute 

Conditions

Threshold type

☒ **Static**
Use a value as a threshold

☐ **Anomaly detection**
Use a band as a threshold

Whenever CPUUtilization is...
Define the alarm condition.

☐ Greater
> threshold

☒ **Greater/Equal**
>= threshold

☐ Lower/Equal
<= threshold

☐ Lower
< threshold

than...
Define the threshold value.

80

Must be a number

CloudShell Feedback © 2025, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

Remove and next

Create alarm | Alarms | CloudWatch | us-east-2 - Work - Microsoft Edge

https://us-east-2.console.aws.amazon.com/cloudwatch/home?region=us-east-2#alarmsV2:create?~(Page~'Actions~'AlarmType~'MetricAlarm~'AlarmData~(Namespace~'AWS*2fEC2~MetricN...

CloudWatch > Alarms > Create alarm

Alarm recommendations available
Turn on Recommendations to pre-populate the wizard with the recommended alarms.

Step 1
Specify metric and conditions

Step 2
Configure actions

Step 3
Add alarm details

Step 4
Preview and create

Configure actions

Notification
[Add notification](#)

Lambda action
[Add Lambda action](#)

Auto Scaling action
[Add Auto Scaling action](#)

CloudShell Feedback © 2025, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences 103°F Haze 4:08 PM 6/8/2025

Create alarm | Alarms | CloudWatch | us-east-2 - Work - Microsoft Edge

https://us-east-2.console.aws.amazon.com/cloudwatch/home?region=us-east-2#alarmsV2:create?~(Page~'Details~'AlarmType~'MetricAlarm~'AlarmData~(Namespace~'AWS*2fEC2~MetricN...

CloudWatch > Alarms > Create alarm

Alarm recommendations available
Turn on Recommendations to pre-populate the wizard with the recommended alarms.

Step 1
Specify metric and conditions

Step 2
Configure actions

Step 3
Add alarm details

Step 4
Preview and create

Add alarm details

Name and description

Alarm name
CPUALert

Alarm description - optional [View formatting guidelines](#)

[Edit](#) [Preview](#)

This is an H1
double asterisks will produce strong character
This is [an example](https://example.com/) inline link.

Up to 1024 characters (0/1024)

Markdown formatting is only applied when viewing your alarm in the console. The description will remain in plain text in the

CloudShell Feedback © 2025, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences 103°F Haze 4:09 PM 6/8/2025

Next -> create alarm

Alarms | CloudWatch | us-east-2 - Work - Microsoft Edge

https://us-east-2.console.aws.amazon.com/cloudwatch/home?region=us-east-2#alarmsV2?~(Page~'MetricSelection~AlarmType~'MetricAlarm~AlarmData~(Metrics~(~)~AlarmName~'~Ala...

CloudWatch > Alarms

Successfully created alarm CPUAlert. View alarm

Alarms (1)

☐ Hide Auto Scaling alarms Clear selection Create composite alarm Actions Create alarm

Search Alarm state: Any Alarm type: Any Actions status: Any < 1 >

☐	Name	State	Last state update (UTC)	Conditions
☐	CPUAlert	Insufficient data	2025-06-08 10:39:43	CPUUtilization >= 80 for 1 datapoints with minute

CloudShell Feedback © 2025, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

103°F Haze Search 4:09 PM 6/8/2025

Task 3: Connect to an EC2 instance by using EC2 Instance Connect

Instances | EC2 | us-east-2 - Work - Microsoft Edge

https://us-east-2.console.aws.amazon.com/ec2/home?region=us-east-2#Instances:

EC2 > Instances

Instances (1/1) Info Last updated less than a minute ago Connect Instance state Actions Launch instances

Find Instance by attribute or tag (case-sensitive) All states < 1 >

☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status
☒	WebServer	i-05a53b9a5f6af914a	Running	t2.micro	2/2 checks passed	View alarms

i-05a53b9a5f6af914a (WebServer)

Details Status and alarms Monitoring Security Networking Storage Tags

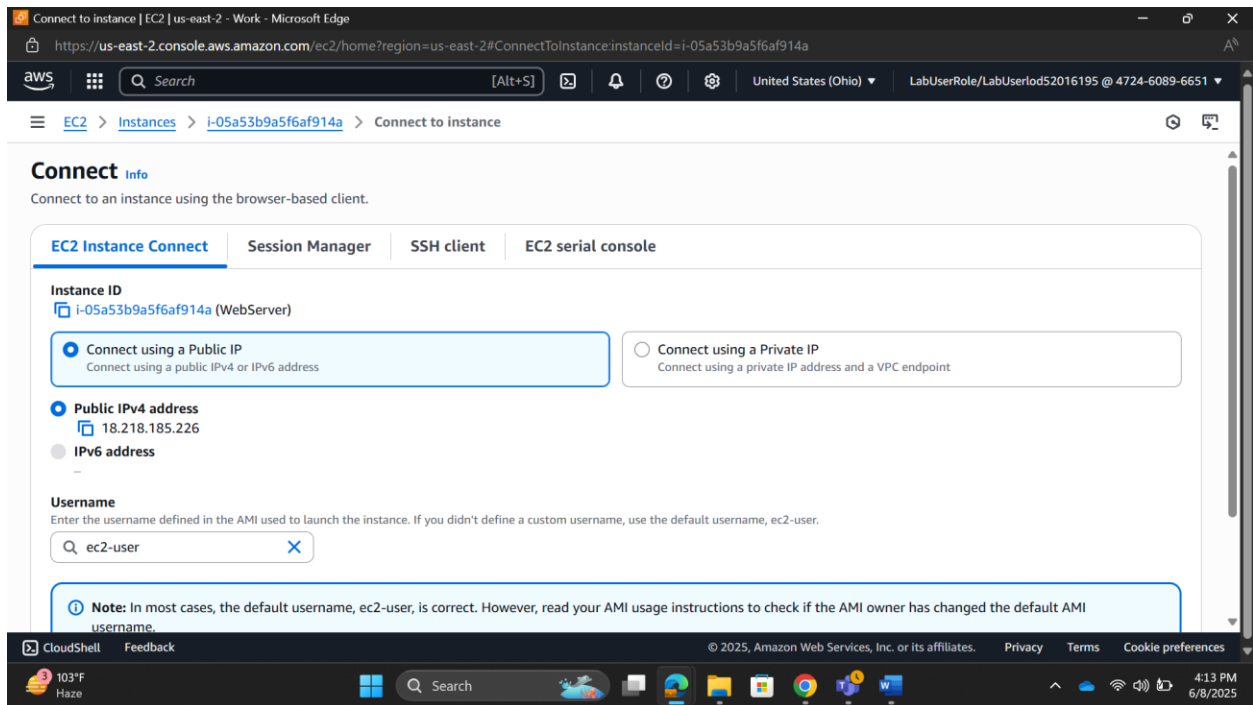
Instance summary Info

Instance ID	Public IPv4 address	Private IPv4 addresses
i-05a53b9a5f6af914a	18.218.185.226 open address	10.0.0.211
IPv6 address	Instance state	Public DNS
	Running	

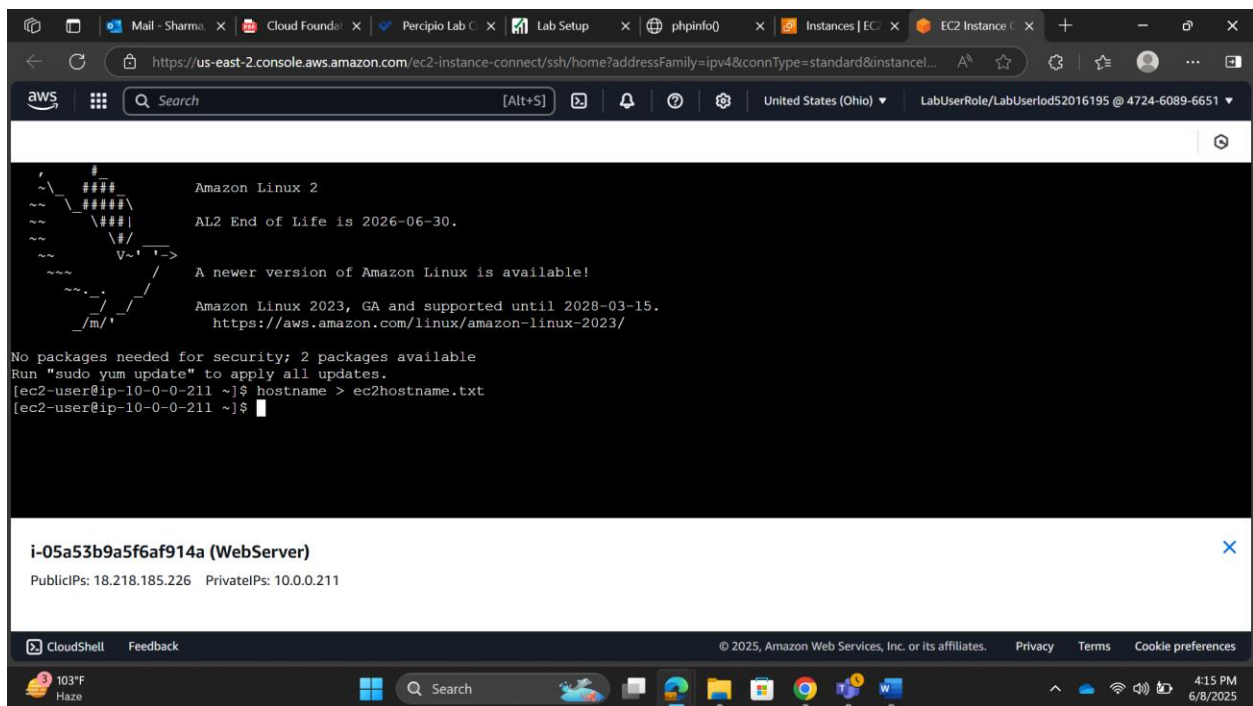
https://us-east-2.console.aws.amazon.com/ec2/home?region=us-east-2#Co... © 2025, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

103°F Haze Search 4:12 PM 6/8/2025

Click connect

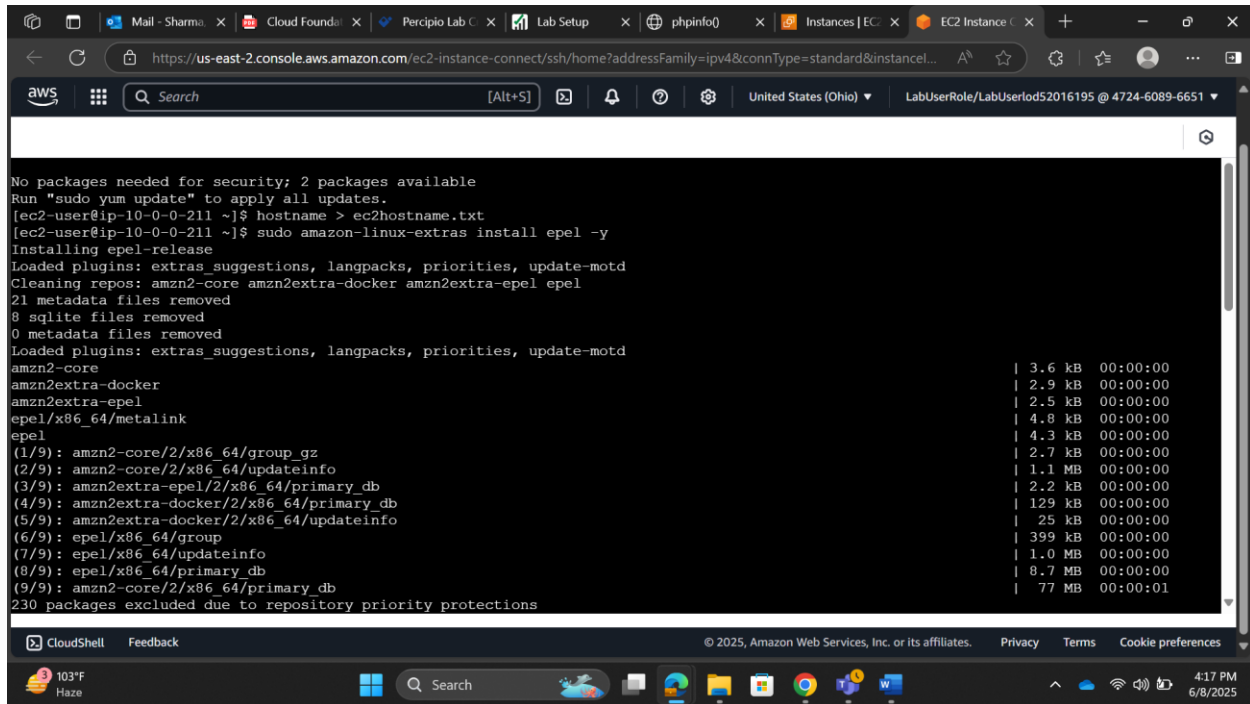


Click connect



Task 4: Install and run the CPU stress tool

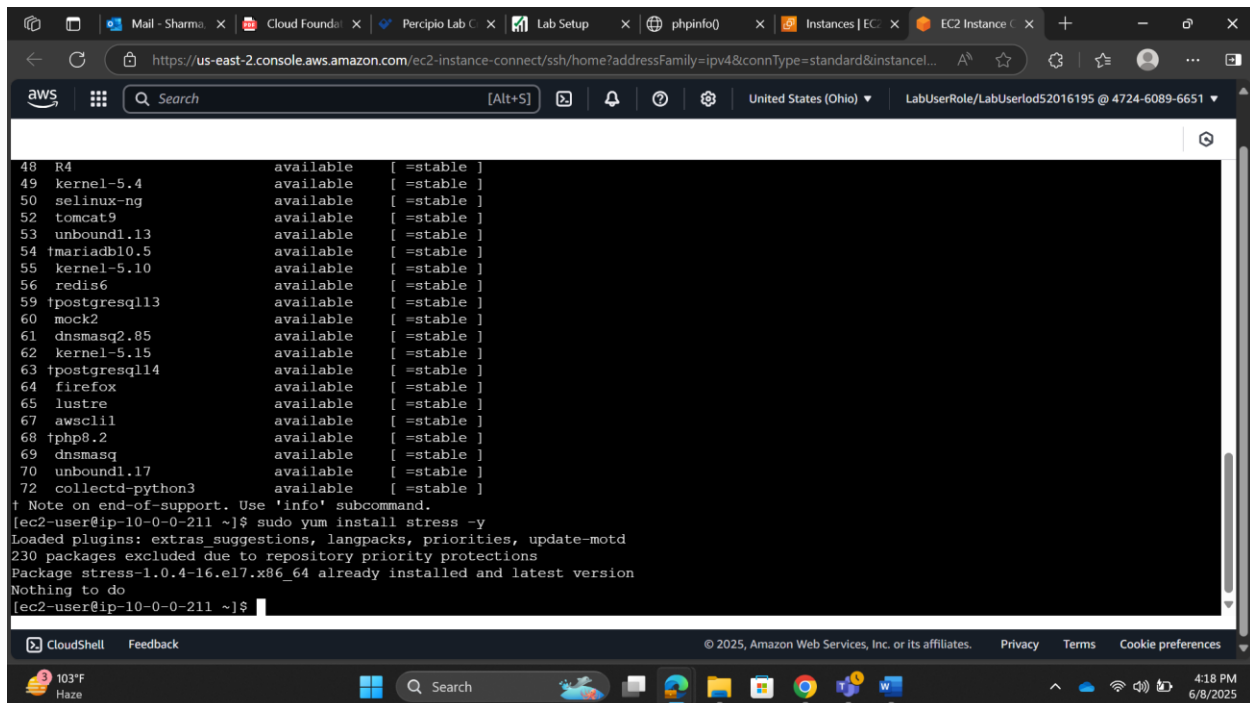
Run this command: `sudo amazon-linux-extras install epel -y`



The screenshot shows a terminal window in the AWS CloudShell interface. The user is logged in as 'ec2-user' on an Amazon Linux 2 instance. The terminal output shows the execution of the command `sudo amazon-linux-extras install epel -y`. The output indicates that 2 packages are available for installation. The packages listed are `amzn2-core`, `amzn2extra-docker`, `amzn2extra-epel`, and `epel`. The installation progress is shown with a table of packages and their sizes. The installation is successful, and the user is prompted to run `sudo yum update` to apply all updates.

```
No packages needed for security; 2 packages available
Run "sudo yum update" to apply all updates.
[ec2-user@ip-10-0-0-211 ~]$ hostname > ec2hostname.txt
[ec2-user@ip-10-0-0-211 ~]$ sudo amazon-linux-extras install epel -y
Installing epel-release
Loaded plugins: extras_suggestions, langpacks, priorities, update-motd
Cleaning repos: amzn2-core amzn2extra-docker amzn2extra-epel epel
21 metadata files removed
8 sqlite files removed
0 metadata files removed
Loaded plugins: extras_suggestions, langpacks, priorities, update-motd
amzn2-core | 3.6 kB 00:00:00
amzn2extra-docker | 2.9 kB 00:00:00
amzn2extra-epel | 2.5 kB 00:00:00
epel/x86_64/metalink | 4.8 kB 00:00:00
epel | 4.3 kB 00:00:00
(1/9): amzn2-core/2/x86_64/group_gz | 2.7 kB 00:00:00
(2/9): amzn2-core/2/x86_64/updateinfo | 1.1 MB 00:00:00
(3/9): amzn2extra-epel/2/x86_64/primary_db | 2.2 kB 00:00:00
(4/9): amzn2extra-docker/2/x86_64/primary_db | 129 kB 00:00:00
(5/9): amzn2extra-docker/2/x86_64/updateinfo | 25 kB 00:00:00
(6/9): epel/x86_64/group | 399 kB 00:00:00
(7/9): epel/x86_64/updateinfo | 1.0 MB 00:00:00
(8/9): epel/x86_64/primary_db | 8.7 MB 00:00:00
(9/9): amzn2-core/2/x86_64/primary_db | 77 MB 00:00:01
230 packages excluded due to repository priority protections
```

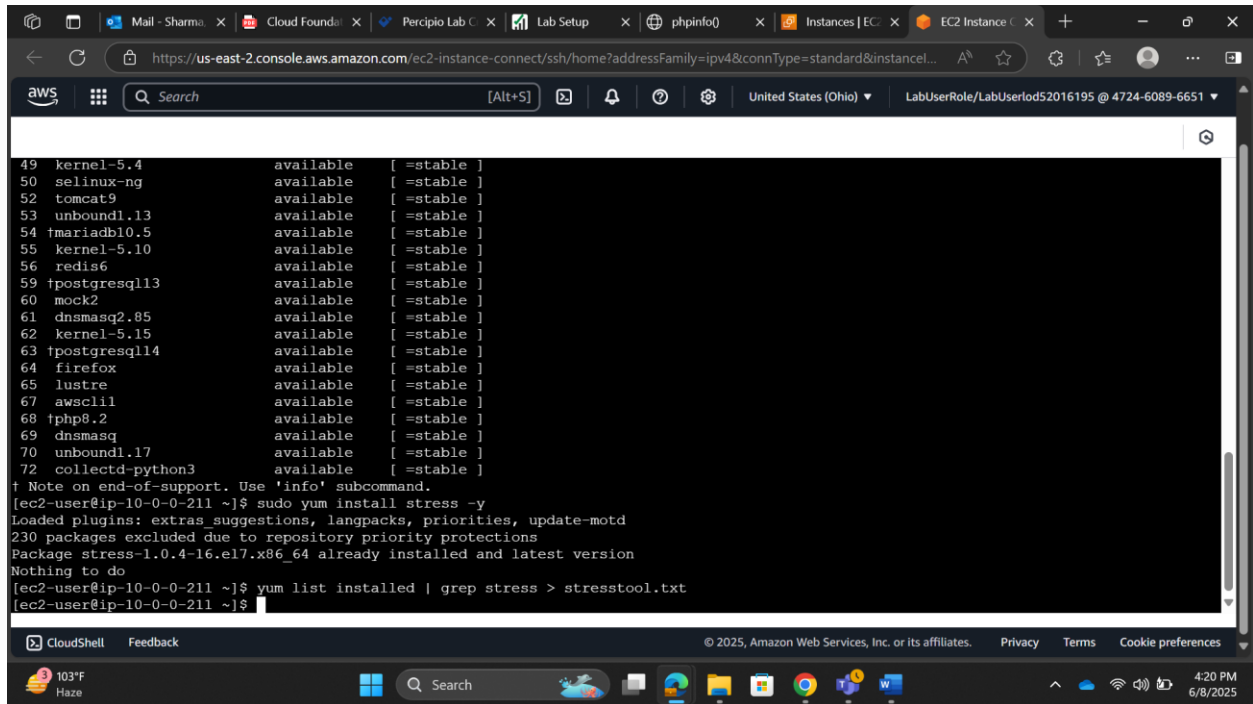
`sudo yum install stress -y`



The screenshot shows a terminal window in the AWS CloudShell interface. The user is logged in as 'ec2-user' on an Amazon Linux 2 instance. The terminal output shows the execution of the command `sudo yum install stress -y`. The output indicates that 230 packages are excluded due to repository priority protections. The package `stress-1.0.4-16.el7.x86_64` is already installed and the latest version is available. The user is prompted to run `sudo yum update` to apply all updates.

```
48 R4 available [ =stable ]
49 kernel-5.4 available [ =stable ]
50 selinux-ng available [ =stable ]
52 tomcat9 available [ =stable ]
53 unbound1.13 available [ =stable ]
54 tmariadb10.5 available [ =stable ]
55 kernel-5.10 available [ =stable ]
56 redis6 available [ =stable ]
59 tpostgresql13 available [ =stable ]
60 mock2 available [ =stable ]
61 dnsmasq2.85 available [ =stable ]
62 kernel-5.15 available [ =stable ]
63 tpostgresql14 available [ =stable ]
64 firefoxf available [ =stable ]
65 lustre available [ =stable ]
67 awsccli1 available [ =stable ]
68 tphp8.2 available [ =stable ]
69 dnsmasq available [ =stable ]
70 unbound1.17 available [ =stable ]
72 collectd-python3 available [ =stable ]
† Note on end-of-support. Use 'info' subcommand.
[ec2-user@ip-10-0-0-211 ~]$ sudo yum install stress -y
Loaded plugins: extras_suggestions, langpacks, priorities, update-motd
230 packages excluded due to repository priority protections
Package stress-1.0.4-16.el7.x86_64 already installed and latest version
Nothing to do
[ec2-user@ip-10-0-0-211 ~]$
```

yum list installed | grep stress > stresstool.txt

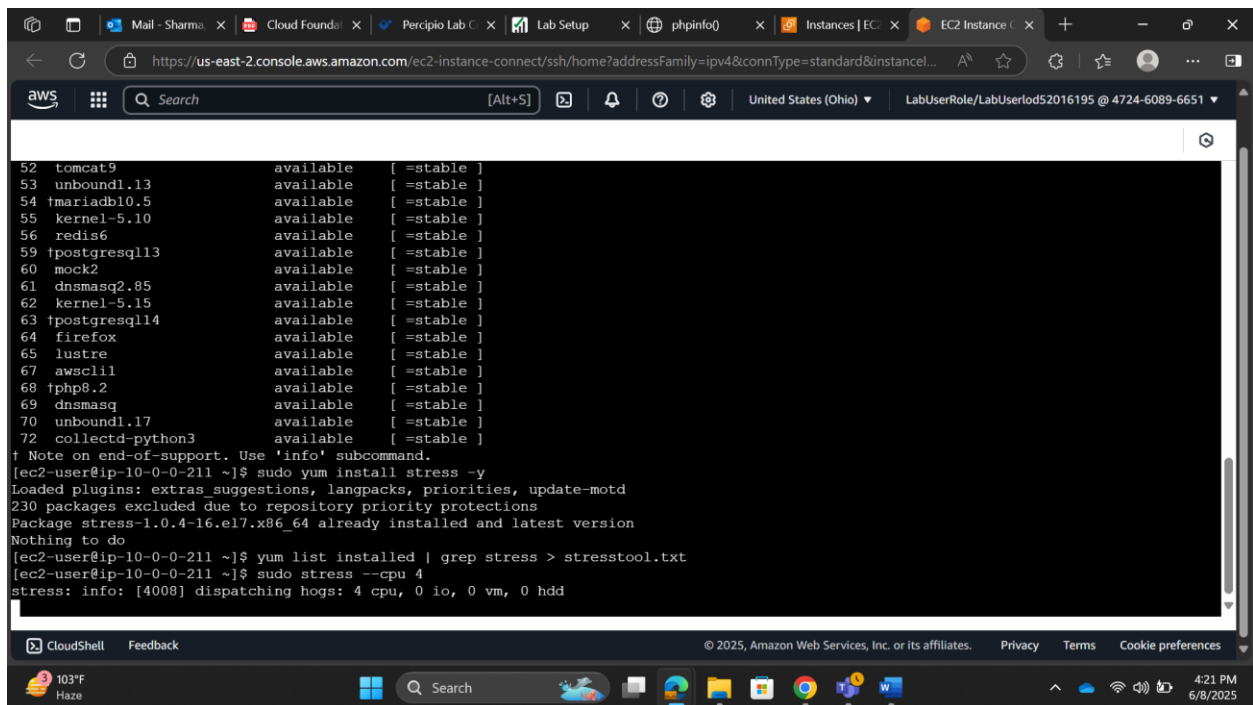


The screenshot shows an AWS CloudShell terminal window. The terminal output lists various installed packages, including kernel-5.4, selinux-ng, tomcat9, unbound1.13, tmariadb10.5, kernel-5.10, redis6, tpostgresql13, mock2, dnsmasq2.85, kernel-5.15, tpostgresql14, firefox, lustre, awscli1, tphp8.2, dnsmasq, unbound1.17, and collectd-python3. The output is formatted as a table with columns for package name, architecture, and version. Below the list, a note mentions end-of-support and the use of the 'info' subcommand. The user then runs 'sudo yum install stress -y', which shows that the stress package is already installed. Finally, the user runs 'yum list installed | grep stress > stresstool.txt' to save the output to a file.

```
49 kernel-5.4                available [ =stable ]
50 selinux-ng                 available [ =stable ]
52 tomcat9                    available [ =stable ]
53 unbound1.13                available [ =stable ]
54 tmariadb10.5               available [ =stable ]
55 kernel-5.10                available [ =stable ]
56 redis6                     available [ =stable ]
59 tpostgresql13              available [ =stable ]
60 mock2                       available [ =stable ]
61 dnsmasq2.85                available [ =stable ]
62 kernel-5.15                available [ =stable ]
63 tpostgresql14              available [ =stable ]
64 firefox                     available [ =stable ]
65 lustre                     available [ =stable ]
67 awscli1                    available [ =stable ]
68 tphp8.2                    available [ =stable ]
69 dnsmasq                     available [ =stable ]
70 unbound1.17                available [ =stable ]
72 collectd-python3           available [ =stable ]

+ Note on end-of-support. Use 'info' subcommand.
[ec2-user@ip-10-0-0-211 ~]$ sudo yum install stress -y
Loaded plugins: extras_suggestions, langpacks, priorities, update-motd
230 packages excluded due to repository priority protections
Package stress-1.0.4-16.el7.x86_64 already installed and latest version
Nothing to do
[ec2-user@ip-10-0-0-211 ~]$ yum list installed | grep stress > stresstool.txt
[ec2-user@ip-10-0-0-211 ~]$
```

sudo stress --cpu 4



The screenshot shows the same AWS CloudShell terminal window. The user runs 'sudo stress --cpu 4', which starts the stress test. The output shows the stress test is running with 4 CPU hogs. The terminal output is as follows:

```
52 tomcat9                    available [ =stable ]
53 unbound1.13                available [ =stable ]
54 tmariadb10.5               available [ =stable ]
55 kernel-5.10                available [ =stable ]
56 redis6                     available [ =stable ]
59 tpostgresql13              available [ =stable ]
60 mock2                       available [ =stable ]
61 dnsmasq2.85                available [ =stable ]
62 kernel-5.15                available [ =stable ]
63 tpostgresql14              available [ =stable ]
64 firefox                     available [ =stable ]
65 lustre                     available [ =stable ]
67 awscli1                    available [ =stable ]
68 tphp8.2                    available [ =stable ]
69 dnsmasq                     available [ =stable ]
70 unbound1.17                available [ =stable ]
72 collectd-python3           available [ =stable ]

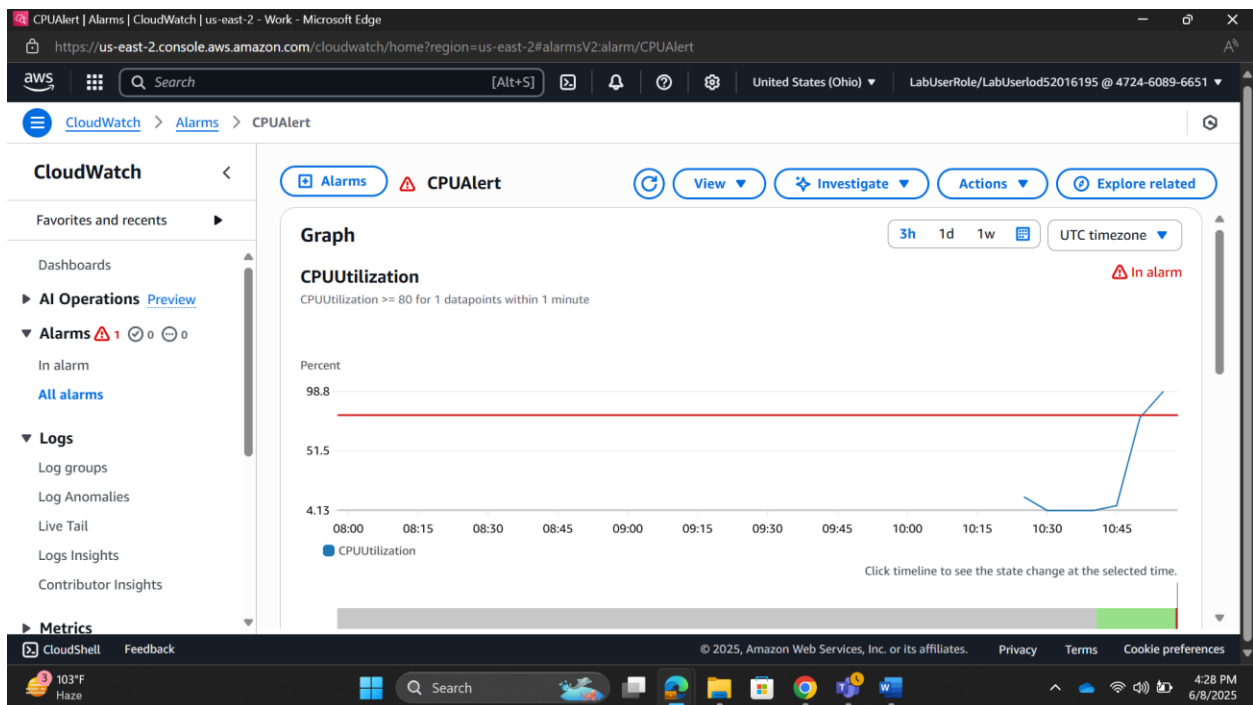
+ Note on end-of-support. Use 'info' subcommand.
[ec2-user@ip-10-0-0-211 ~]$ sudo yum install stress -y
Loaded plugins: extras_suggestions, langpacks, priorities, update-motd
230 packages excluded due to repository priority protections
Package stress-1.0.4-16.el7.x86_64 already installed and latest version
Nothing to do
[ec2-user@ip-10-0-0-211 ~]$ yum list installed | grep stress > stresstool.txt
[ec2-user@ip-10-0-0-211 ~]$ sudo stress --cpu 4
stress: info: [4008] dispatching hogs: 4 cpu, 0 io, 0 vm, 0 hdd
```

Task 5: Monitor the CPUAlert alarm for the WebServer EC2 instance

The screenshot shows the AWS CloudWatch Alarms console. The left sidebar contains navigation links for CloudWatch, Alarms, Logs, and Metrics. The main area displays a list of alarms. The 'CPUAlert' alarm is shown with a status of 'OK'. The conditions for the alarm are 'CPUUtilization >= 80 for 1 datapoints within 1 minute'. The last state update is '2025-06-08 10:40:41'.

Name	State	Last state update (UTC)	Conditions
CPUAlert	OK	2025-06-08 10:40:41	CPUUtilization >= 80 for 1 datapoints within 1 minute

Periodically refresh the CPUAlert pane until the CPU utilization exceeds the 80 percent threshold in the alarm and the status changes to **In Alarm**.



labclient.labondemand.com/Instructions/ExamResult/d057d90a-54d8-407d-847b-a47f9d97c16b - Work - Microsoft Edge

https://labclient.labondemand.com/Instructions/ExamResult/d057d90a-54d8-407d-847b-a47f9d97c16b

Great job! [Print](#)

Your dedication and hard work really paid off!



Areas where you excelled

- **Connect to an EC2 instance**
You completed this area with great success.
- **Create a CloudWatch alarm to monitor CPU utilization**
After completing this section, it's obvious you know what you're talking about.
- **Connect to an EC2 instance by using EC2 Instance Connect**
You showed your skills in this area and finished with success.

Hot days ahead
103°F

Search

4:30 PM
6/8/2025

You are done with the lab!!!